Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
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## 1. Scope and method

Screening only — not a piping stress analysis. Thermal growth of a straight run between anchors, the fully restrained stress and force, the leg an offset leg or a U-loop (two guided cantilevers) needs to absorb the growth within the owner's allowable displacement-stress range Sa and whether the available leg is long enough (an offset leg or Z-bend also loads its run or runs with the leg's own growth), the force on the anchors and the largest root moment of the guided-cantilever model, the screening ratio $D \cdot Y/(L - U)^2$ against the owner's K1 (PASS, or the line needs a formal stress analysis), and, for an axial bellows, the anchor force P·A_eff plus its spring force. No allowable stress, K1 or code table ships.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Pipe section | $d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$ |
| Temperature change | $\Delta T = T_{operate} - T_{install}$, or as entered; the movement uses $\lvert\Delta T\rvert$ and its sign gives the direction |
| Run that grows | $L_{tot} = L$, or $L + L2$ for a Z-bend; thermal growth $\Delta L = \alpha \cdot L_{tot} \cdot \lvert\Delta T\rvert$ |
| Fully restrained pipe | $\sigma_r = E \cdot \alpha \cdot \lvert\Delta T\rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert\Delta T\rvert \cdot A_m$ ($E_r = E_{hot}$, else E) |
| Movement one leg must absorb | $\Delta = \Delta L$ for an offset leg or Z-bend; $\Delta = \Delta L/2$ for a U-loop (two identical legs in series, the cross leg a rigid guide) |
| Guided cantilever, required leg | $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta/Sa)}$, the leg at which the bending stress M/Z reaches Sa; PASS when $L_{req} \le$ the available leg H |
| Force and moment of the available leg | $F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta/H^3$ (acts along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta/H^2$ ($= F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta/H^2 = Sa \cdot (L_{req}/H)^2$ |
| Offset leg or Z-bend, the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert\Delta T\rvert \cdot H$ | the runs act in series under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg}/(L^3 + L2^3)$ ($L2 = 0$ for an L-bend) acting along the leg; run i takes the share $L_i^3/(L^3 + L2^3)$ of $\Delta_{leg}$, with root stress $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i/L_i^2$ and moment $M_i = F_{run} \cdot L_i/2$; each run is checked as its own cantilever, PASS when $\sqrt{(\sigma_i/Sa)} \le 1$ (for an L-bend the run needed is $L_{req,run} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta_{leg}/Sa)}$) |
| Force on the anchors | the two components are perpendicular, $F_{leg}$ along the run and $F_{run}$ along the leg, so the force is $\sqrt{(F_{leg}^2 + F_{run}^2)}$; the moment reported is the largest root moment of the legs and runs in the guided-cantilever model |
| Flexibility screening | $OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert\Delta T\rvert \cdot U$, the free growth between the anchors; offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$ (so $Y = \Delta L$); PASS, or the line needs a formal stress analysis |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Axial bellows (no leg or loop, no leg check, no screening) | anchor force = $P \cdot A_{eff} + k \cdot \Delta L$; the movement is checked against the bellows' rated axial movement when given |

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](para. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](para. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| E | Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa) | 120 | GPa | assumption: W3 seed 'Copper (generic)', indicative; replace with the project's material record. | 0.5 ... 500 GPa |
| α | Mean coefficient of thermal expansion between the install and operating temperatures | 17 | um/(m·K) | assumption: W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record. | 1 ... 250 um/(m·K) |
| OD | Pipe outside diameter | 54 | mm | drawing: DRG-M-212 rev A | 10 ... 1500 mm |
| $t_{wall}$ | Pipe wall thickness | 1.5 | mm | drawing: DRG-M-212 rev A | 0.5 ... 60 mm |
| L | Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth) | 24 | m | drawing: DRG-M-212 rev A | 1 ... 500 m |
| $temp_{mode}$ | Temperature change from | install and operating temperatures | | assumption: Install and operating temperatures are both known. | any |
| $T_{install}$ | Temperature at installation (when the run is anchored) | 10 | °C | assumption: Mean site temperature when the riser is anchored. | -50 ... 50 °C |
| $T_{operate}$ | Operating temperature (hot or cold) | 70 | °C | spec: SPEC-M-04 §3: heating flow temperature | -50 ... 500 °C |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| layout | What absorbs the movement | offset leg (L- or Z-bend: the leg takes the whole movement) | | drawing: DRG-M-212 rev A | any |
| H | Available leg length (each leg of a U-loop; an offset leg: the leg off the run, which may be the longer of the two); no default, not needed for an axial bellows | 2.5 | m | drawing: DRG-M-212 rev A | 0.5 … 50 m |
| Sa | Allowable displacement-stress range Sa for the legs and runs (your piping code and its formula; no default; not needed for an axial bellows) | 60 | MPa | assumption: Invented illustrative allowable range; take Sa from the project's piping code and its formula. | 5 … 500 MPa |
| $K_{scr}$ | Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², …) from your piping code (no default; blank = screening not evaluated and the sheet does not pass) | 100 | mm²/m² | assumption: Invented illustrative screening limit; take K1 from the project's piping code. | 1 … 1000 mm²/m² |

## 3. Assumptions

A1. Screening only — not a piping stress analysis. The result tells whether a simple leg or loop is plausible and what it puts on its anchors; it does not replace a flexibility analysis of the real routing (elbow flexibility, intensification, torsion, branches, restraint friction, supports, sustained and occasional loads). (Stated scope.)

A2. A guided cantilever: the available leg is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs (stiffness $12 \cdot E \cdot I / H^3$, end moment $F \cdot H/2$). Elbows are taken as rigid and the leg as slender: the beam model is not valid for legs shorter than about ten diameters. (Beam theory.)

A3. No stress-intensification factor and no elbow flexibility are applied: the stress at the elbow root of a real bend is higher than the beam stress M/Z, by the intensification factor of the code, so a leg that passes here can still exceed Sa at the elbow. This is the main reason the result is a screen and not a stress analysis. (Stated limit.)

A4. A U-loop is two identical legs in series, each a guided cantilever taking half the growth, the cross leg a rigid guide; the force along the line is the same F in both legs. The moment M is the moment at the leg roots (the corners of the loop); it reaches an anchor only where the leg ends at the anchor (an offset leg). The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)

A5. An offset leg is anchored at both ends: the run L bends for the growth of the leg ($\alpha \cdot |\Delta T| \cdot H$) as a second guided cantilever and is checked against the same Sa. A Z-bend (second run L2) has the two runs in series under one force, each taking its share of the leg's growth in proportion to its flexibility, $L_i^3$, and each checked as its own cantilever; the leg takes the growth of both runs. The force on the anchors is the resultant of the force along the run (the leg's) and the force along the leg (the runs'), which are perpendicular. (Offset-leg and Z-bend model.)

A6. $anchor_{moment}$ is the largest root moment of the legs and runs in the guided-cantilever simplification

(M = F·L/2 with each end fixed against rotation). It is not the moment on a frame anchor, which depends on the whole frame (the stiffness of the other legs, the elbow rotations, the anchor's own fixity); a formal analysis is needed before an anchor is sized for moment. (Stated limit.)
A7. The material properties are single values: α the mean coefficient between the install and operating temperatures; E the modulus at the installation (reference) temperature, used for the stress check, as the piping codes do (ASME B31.3 para. 319.3.2 and 319.5, numbers to be checked against the owner's copy); $E_{hot}$, when given, is used only for the reactions on the anchors. Take them from a W3 material applied at the right temperature. (Material library; code convention to be checked.)
A8. Sa and K1 are the owner's records from the owner's own copy of the piping code (or the design basis), with their provenance; no allowable stress, K1 or code constant ships. The sheet applies the relation to the numbers entered and does not know which code, material or load case they belong to. (No table from a standard is shipped.)
A9. No friction, weight, spring-hanger or cold-spring effect is included in the leg force; the anchor force adds the friction of the sliding supports, μ·w·L, only when μ and the weight per metre are entered (a warning says so when they are not). The axial bellows anchor force is the pressure thrust $P \cdot A_{eff}$ plus the spring force k·ΔL at the governing pressure entered (the design or the test pressure, whichever governs; a bellows' compression and extension ratings can differ): guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A10. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 120 GPa. (W3 seed 'Copper (generic)', indicative; replace with the project's material record.)
A11. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 17 um/(m*K). (W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.)
A12. $temp_{mode}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A13. $T_{install}$ (Temperature at installation (when the run is anchored)) is taken as 10 degC. (Mean site temperature when the riser is anchored.)
A14. Sa (Allowable displacement-stress range Sa for the legs and runs (your piping code and its formula; no default; not needed for an axial bellows)) is taken as 60 MPa. (Invented illustrative allowable range; take Sa from the project's piping code and its formula.)
A15. $K_{scr}$ (Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², ...) from your piping code (no default; blank = screening not evaluated and the sheet does not pass)) is taken as 100 mm^2/m^2. (Invented illustrative screening limit; take K1 from the project's piping code.)
A16. Layout: the offset leg of 2.5 m takes the whole movement of 24.48 mm as a guided cantilever fixed against rotation at both ends, and the run of 24 m bends for the leg's growth of 2.55 mm. Screening only — not a piping stress analysis. (Guided-cantilever model.)
A17. The available leg (2.5 m) is shorter than the 2.8163 m required: the bending stress at its root is 76.143 MPa against Sa = 60 MPa. Lengthen the leg or loop, add a loop or bellows, or analyse the line formally. (Result of the check.)

## 4. Working

[9] $dT = T_{operate} - T_{install}$
$= 70\ degC - 10\ degC$
$= 60\ K$
Temperature change, operating – install

[11] $E_r = E$
$= 120\ GPa$
$= 120\ GPa$
Modulus for the reactions (E, no $E_{hot}$ given)

[12] $D_i$ = OD - 2 * $t_{wall}$
= 54 mm − 2 × 1.5 mm
= 51 mm
Inside diameter
[13] $A_m$ = pi / 4 * (OD^2 - $D_i$^2)
= π / 4 × ((54 mm)^2 − (51 mm)^2)
= 247.4 mm²
Metal cross-section
[14] $I_x$ = pi / 64 * (OD^4 - $D_i$^4)
= π / 64 × ((54 mm)^4 − (51 mm)^4)
= 8.5307 cm⁴
Second moment of area of the pipe section
[15] $Z_x$ = $I_x$ / (OD / 2)
= 8.5307 cm⁴ / (54 mm / 2)
= 3.1595 cm³
Section modulus
[16] $L_{tot}$ = L
= 24 m
= 24 m
Length that grows between the anchors
[17] dL = α * $L_{tot}$ * abs(dT)
= 17 um/(m·K) × 24 m × abs(60 K)
= 24.48 mm
Thermal growth between the anchors, as a magnitude
[18] $dL_{signed}$ = α * $L_{tot}$ * dT
= 17 um/(m·K) × 24 m × 60 K
= 24.48 mm
Thermal movement with its sign: positive lengthens, negative shortens
[19] $\sigma_{res}$ = E * α * abs(dT)
= 120 GPa × 17 um/(m·K) × abs(60 K)
= 122.4 MPa
Stress in the fully restrained pipe (E at the installation temperature)
[20] $F_{res}$ = $E_r$ * α * abs(dT) * $A_m$
= 120 GPa × 17 um/(m·K) × abs(60 K) × 247.4 mm²
= 30.282 kN
Force on the anchors of the fully restrained pipe
[21] $F_{friction}$ = support friction not entered
= μ and $w_{pipe}$ blank
= 0 N
Friction of the sliding supports (not entered: none in the anchor force)
[24] $u_{res}$ = $\sigma_{res}$ / Sa
= 122.4 MPa / 60 MPa
= 2.04
Fully restrained stress as a fraction of Sa (above 1: the movement has to be absorbed)
[25] Δ = dL
= 24.48 mm
= 24.48 mm
Movement the offset leg absorbs (all of it)
[26] $L_{req}$ = sqrt(3 * E * OD * Δ / Sa)
= sqrt(3 × 120 GPa × 54 mm × 24.48 mm / 60 MPa)
= 2.8163 m
Leg length at which the guided-cantilever bending stress reaches Sa

[27] $u_{leg} = L_{req} / H$
$= 2.8163\ m / 2.5\ m$
$= 1.1265$
Required leg over available leg
[28] $F_{leg} = 12 * E_r * I_x * \Delta / H^3$
$= 12 \times 120\ GPa \times 8.5307\ cm^4 \times 24.48\ mm / (2.5\ m)^3$
$= 192.46\ N$
Force of the available leg on its anchors or guides
[29] $M_{leg} = 6 * E_r * I_x * \Delta / H^2$
$= 6 \times 120\ GPa \times 8.5307\ cm^4 \times 24.48\ mm / (2.5\ m)^2$
$= 240.57\ N·m$
Moment at the root of the available leg
[30] $\sigma_{leg} = 3 * E * OD * \Delta / H^2$
$= 3 \times 120\ GPa \times 54\ mm \times 24.48\ mm / (2.5\ m)^2$
$= 76.143\ MPa$
Bending stress at the root of the available leg (E at the installation temperature)
[31] $u_{sigma} = \sigma_{leg} / Sa$
$= 76.143\ MPa / 60\ MPa$
$= 1.269$
Bending stress over Sa (= $u_{leg}^2$)
[32] $\Delta_{run} = \alpha * abs(dT) * H$
$= 17\ um/(m·K) \times abs(60\ K) \times 2.5\ m$
$= 2.55\ mm$
Growth of the leg, which the run or runs have to absorb
[33] $S_{run} = L^3$
$= (24\ m)^3$
$= 13824\ m^3$
Cube of the run (its flexibility)
[34] $F_{run} = 12 * E_r * I_x * \Delta_{run} / S_{run}$
$= 12 \times 120\ GPa \times 8.5307\ cm^4 \times 2.55\ mm / 13824\ m^3$
$= 0.02266\ N$
Force of the run or runs, along the leg
[35] $\sigma_{run} = 3 * E * OD * \Delta_{run} * L / S_{run}$
$= 3 \times 120\ GPa \times 54\ mm \times 2.55\ mm \times 24\ m / 13824\ m^3$
$= 0.086062\ MPa$
Bending stress at the root of the run
[36] $u_{run} = sqrt(\sigma_{run} / Sa)$
$= sqrt(0.086062\ MPa / 60\ MPa)$
$= 0.037873$
Required run over available run (= √(σ/Sa))
[37] $L_{req,run} = L * u_{run}$
$= 24\ m \times 0.037873$
$= 0.90896\ m$
Cantilever length for this run's share of the leg's growth
[38] $M_{run} = F_{run} * L / 2$
$= 0.02266\ N \times 24\ m / 2$
$= 0.27192\ N·m$
Moment at the root of the run
[39] $F_{along,run} = F_{leg} + F_{friction}$
$= 192.46\ N + 0\ N$
$= 192.46\ N$
Component of the anchor force along the run: the leg's force plus the support friction

[40] $F_{along,leg} = F_{run}$
= 0.02266 N
= 0.02266 N
Component of the anchor force along the leg: the runs' force
[41] $anchor_{force} = sqrt(F_{along,run}^2 + F_{run}^2)$
= sqrt((192.46 N)^2 + (0.02266 N)^2)
= 192.46 N
Force on the anchors: resultant of the two perpendicular components (support friction only if μ and $w_{pipe}$ are entered)
[42] $anchor_{moment} = max(M_{leg}, M_{run})$
= max(240.57 N·m, 0.27192 N·m)
= 240.57 N·m
Largest root moment of the leg and the run or runs (guided-cantilever model)
[43] $L_{dev} = L_{tot} + H$
= 24 m + 2.5 m
= 26.5 m
Developed length between the anchors, run or runs plus leg
[44] $U_a = sqrt(L_{tot}^2 + H^2)$
= sqrt((24 m)^2 + (2.5 m)^2)
= 24.13 m
Straight distance between the anchors
[45] $Y_{scr} = \alpha * abs(dT) * U_a$
= 17 um/(m·K) × abs(60 K) × 24.13 m
= 24.612 mm
Resultant displacement Y: the free growth between the anchors
[46] $scr = OD * Y_{scr} / (L_{dev} - U_a)^2$
= 54 mm × 24.612 mm / (26.5 m − 24.13 m)^2
= 236.59 mm²/m²
Screening ratio OD·Y/(L − U)²
[48] $u_{scr} = scr / K_{scr}$
= 236.59 mm²/m² / 100 mm²/m²
= 2.3659
Screening ratio over the K1 record
[49] pass = K1 given and every check passes
= 3 check(s)
= 0
1 when the leg, the run (offset leg) and the screening checks all pass; 0 when one fails or when no K1 was entered

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $D_i$ | 51 mm | Inside diameter |
| $A_m$ | 247.4 mm² | Metal cross-section |
| $I_x$ | 8.5307 cm⁴ | Second moment of area |
| $Z_x$ | 3.1595 cm³ | Section modulus |
| dT | 60 K | Temperature change |
| $L_{tot}$ | 24 m | Length that grows |
| dL | 24.48 mm | Thermal movement between the anchors (magnitude) |
| $dL_{signed}$ | 24.48 mm | Thermal movement, signed |
| $\sigma_{res}$ | 122.4 MPa | Fully restrained stress |
| $F_{res}$ | 30.282 kN | Fully restrained force |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| direction | "growth: the pipe lengthens; fully restrained it presses on its anchors (compression)" | Direction of the movement |
| $F_{friction}$ | 0 N | Support friction on the run |
| $u_{res}$ | 2.04 | Fully restrained stress over Sa |
| Δ | 24.48 mm | Movement absorbed by each leg |
| $L_{req}$ | 2.8163 m | Required leg length |
| $u_{leg}$ | 1.1265 | Required leg over available leg |
| $F_{leg}$ | 192.46 N | Force of the leg |
| $M_{leg}$ | 240.57 N·m | Moment at the leg root |
| $σ_{leg}$ | 76.143 MPa | Bending stress at the leg root |
| $u_{sigma}$ | 1.269 | Bending stress over Sa |
| $Δ_{run}$ | 2.55 mm | Leg growth absorbed by the run or runs |
| $S_{run}$ | 13824 m³ | Flexibility of the runs |
| $F_{run}$ | 0.02266 N | Force of the runs |
| $σ_{run}$ | 0.086062 MPa | Bending stress at the first run root |
| $L_{req,run}$ | 0.90896 m | Required length of the first run |
| $u_{run}$ | 0.037873 | First run: required over available |
| $M_{run}$ | 0.27192 N·m | Moment at the first run root |
| $F_{along,run}$ | 192.46 N | Anchor force component along the run |
| $F_{along,leg}$ | 0.02266 N | Anchor force component along the leg |
| $anchor_{force}$ | 192.46 N | Force on the anchors (resultant) |
| $anchor_{moment}$ | 240.57 N·m | Largest root moment (guided-cantilever model) |
| $L_{dev}$ | 26.5 m | Developed length |
| $U_a$ | 24.13 m | Anchor distance |
| $Y_{scr}$ | 24.612 mm | Resultant displacement Y |
| scr | 236.59 mm²/m² | Screening ratio OD·Y/(L − U)² |
| $u_{scr}$ | 2.3659 | Screening ratio over K1 |
| screen | "needs formal stress analysis" | Screening result |
| pass | 0 | All checks pass |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Screening only: $OD \cdot Y/(L - U)^2 \le K1$ (not met: the line needs a formal stress analysis) | $u_{scr} = 2.3659$ | $u_{scr} \le 1$ | 2.37 | [1](para. 319.4.1 (screening form; number to be checked against the owner's copy)) |
| FAIL | Available leg against the required leg (2.8163 m) | $u_{leg} = 1.1265$ | $u_{leg} \le 1$ | 1.13 | |
| PASS | Run against the length it needs to absorb the leg's growth (0.90896 m) | $u_{run} = 0.037873$ | $u_{run} \le 1$ | 0.0379 | |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, α (Material properties); OD, $t_{wall}$ (Pipe outside diameter and wall); Sa (Allowable displacement-stress range); $K_{scr}$ (Screening limit K1). Enter each as one so its source, edition and licence travel with it.
WARNING — friction: μ: the anchor force holds no support friction. Every slide or guide between the anchor and the loop adds μ·w·L to it, which on a long run can exceed the leg force several times; enter μ and the weight per metre (with the length on slides) to include it.

## 8. Verification

```
Template bs.pipe-expansion@1.3.0 Content hash 8dd3b0517fce49e5 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. ASME B31.3.
2. EN 13480-3.